

Crosshole ground-penetrating radar full-waveform inversion by combining optimal-transport and least-squares distances

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ABSTRACT

Gradient-based full-waveform inversion using the least-squares distance converges to a local minimum if the starting model is not close enough to the global minimum. In this article, we propose a full-waveform inversion algorithm for crosshole ground penetrating radar tomography that is independent of the starting model. This is achieved by using an optimal-transport distance, which features one broad global minimum and almost no local minima for common ground penetrating radar full-waveform inversion problems. As calculating an optimal-transport distance is computationally expensive and as the broad global minimum of the optimal-transport distance is impacted by numerical approximations in the vicinity of the optimal solution, we propose an algorithm that uses the optimal-transport distance only in the first few iterations of the inversion and switches to the least-squares distance once a solution close enough to the true solution has been found. An additional feature of the proposed algorithm is the sparse and explicit calculation of the gradient using random sets of master points and subsequent interpolation. This avoids the high sensitivity close to the antenna locations, which prohibits meaningful model updates in these regions. Furthermore, this approach smooths the gradient and, thus, the model without an explicit model-regularization term in the objective function. Finally, calculating the gradient explicitly allows to easily implement other distance measures as no update of the gradient calculation becomes necessary. As the explicit calculation of the gradient can be done in parallel and only a limited number of master points is necessary, the additional computational cost is limited. We demonstrate the capabilities of the algorithm on a simple synthetic dataset showing that the proposed algorithm works much better on our example dataset than if only least-squares or optimal-transport distances are used.

1. Introduction

Full-waveform inversion (FWI) [Tarantola, 1984] uses, in contrast to travel-time inversion, significant portions of the complete waveform, either seismic or electromagnetic, to obtain a highly resolved image of the subsurface. An overview of FWI is given by Virieux et al. [2017]. Here we focus on crosshole FWI for ground penetrating radar (GPR), which was first developed in 2007 by two groups independently of each other [Ernst et al., 2007; Kuroda et al., 2007]. One of the main challenges in deterministic least-squares FWI is the strong dependence on the starting model. If the synthetic data based on the starting model differ by more than half a period from the observed data, the algorithm will converge to a local minimum. This problem is known as cycle skipping [Virieux and Operto, 2009].

The standard approach to avoid cycle skipping consists of first performing a travel-time inversion and then initiating the FWI using the

corresponding result [e.g., Klotzsche et al., 2012]. If the result of the travel-time inversion still incurs cycle skipping, one can consider to start the FWI with only the lower frequencies of the data and subsequently add the higher frequency content [e.g., Meles et al., 2012]. In this case, the period at the start of the process is longer than when the complete data are taken into account and, thus, the starting model can be further away from the global minimum. The requirement for successfully using this approach is the presence of low enough frequencies in the data. In the absence of low frequencies, one can also do a Laplace-domain waveform inversion instead of the travel-time inversion to obtain a starting model for FWI [e.g., Shin and Cha, 2008; Bae et al., 2012; Ha and Shin, 2012]. Cycle skipping can also be avoided by using a stochastic inversion approach that can leave a local minimum. However, to make this approach computationally feasible, the dimensionality of the problem, in other words the number of model parameters, needs to be reduced using some sort of model compression while maintaining

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sufficient expressibility of possible models [e.g., Hunziker et al., 2019].

Here we follow another approach, which is to replace the commonly used least-squares (LS) distance to compare waveforms by an optimal-transport (OT) distance, originally formulated by Monge [1781]. Conceptually, the OT distance can be understood by comparing two piles of sand. The order-1 OT distance is the minimum number of shovels of sand that needs to be transported over an as short as possible distance in order to turn one pile of sand into the other. Therefore, this type of optimal-transport distance is also called the Earth-mover distance. Engquist and Froese [2014] demonstrated that this type of distance measure, also known as the Wasserstein distance, does not feature local minima in typical geophysical problems and is, thus, better suited to guide a gradient-based FWI algorithm than the LS distance. The OT distance has been successfully implemented for seismic problems [Métivier et al., 2016, 2018]. For electromagnetic waves, Deng et al. [2022] propose an OT algorithm for asteroid tomography, but a corresponding algorithm for crosshole GPR FWI has, to the authors' knowledge, not yet been developed. It is the goal of this paper to fill that gap and make the algorithm publicly available.

Calculating any type of OT distance between two GPR traces is not straightforward, as it requires the two compared signals to be positive and to contain the same amount of mass [Engquist and Froese, 2014]. As GPR traces oscillate around zero, they contain negative as well as positive values. Furthermore, the energy of one trace might be very different than for another trace, which means that there is no conservation of "mass" between any two traces. To overcome these issues, we follow the approach by Sambridge et al. [2022] and convert each trace into a two-dimensional probability density function, which is positive and has a total mass of one. Since OT distances in general are much simpler to calculate for one-dimensional signals than for two-dimensional ones, Sambridge et al. [2022] propose to compute two marginal probability density functions from this two-dimensional probability density function. This approach by Sambridge et al. [2022] developed for seismics lies at the core of our crosshole GPR OT FWI algorithm.

Calculating the OT distance is computationally much more expensive than calculating the LS distance and uncertainty quantification is challenging as well, because the propagation of uncertainty through the different steps of the calculation of the OT distance is not straightforward. Furthermore, the OT minimum is less well defined than the LS minimum, in the sense that the LS minimum is much narrower as illustrated by Engquist and Froese [2014] and Sambridge et al. [2022] for a simple example.

We propose an algorithm that uses the OT distance only in the first few iterations of the inversion until the resulting model has been improved sufficiently, such that a LS inversion converges to the global minimum. As cycle skipping using the LS distance occurs when the forward-simulated data differ by more than half a period from the observed data, we switch from the OT distance to the LS distance when a critical fraction of the synthetic traces differs by less than half a period from the observed traces. An important aspect is, thus, to determine which fraction of traces needs to fulfill this criterion such that a LS FWI can successfully converge to the global minimum. A similar two-stage approach is often used to invert tracer-test data [e.g., Linde et al., 2006].

Another challenge in gradient-based GPR crosshole FWI is the estimation of model parameters located close to the source and receiver antennas. Most studies do not update the model parameters (dielectric permittivity and conductivity) close to the boreholes. Instead, they keep the starting model inside a muted domain at these locations. Examples include Fig. 9 in Klotzsche et al. [2013] and Fig. 5 in Zhou et al. [2020]. The reason is that the Jacobian (i.e., the derivatives of the simulated data with respect to the model parameters) or the gradient (i.e., the

derivatives of the objective function with respect to the model parameters) are extremely large, even a singularity, close to the sources and receivers. Therefore, without ignoring the Jacobian or the gradient at these locations, the model update occurs primarily close to the antenna locations instead of updating the entire model domain. One solution to this problem is to use a gradient preconditioner as described for seismic data by Kurzmann et al. [2013] and applied to GPR data by van der Kruk et al. [2015]. This allows to decrease the muted domain on each side of the boreholes from 7 model cells to 3 model cells [van der Kruk et al., 2015] or to even eliminate it completely [Klotzsche et al., 2019]. We propose an alternative solution to this problem that uses a numerical calculation of the gradient based on master points followed by an interpolation, inspired by comparable applications in hydrogeology [e.g., RamaRao et al., 1995; Alcolea et al., 2006]. Besides circumventing the high sensitivities around the antennas, this explicit approach also provides a flexible framework to test various distance measures and an intrinsic model regularization.

To update the models iteratively, we use the stochastic gradient descent algorithm, which dates back to the middle of the previous century [Robbins and Monro, 1951] and was used in seismic FWI, for example, by van Leeuwen et al. [2011] or by Yang et al. [2018]. The basic idea of stochastic gradient descent is to use, at each iteration, a different, randomly drawn subset of datapoints instead of the complete dataset to calculate the objective function and the corresponding gradient. In this way, the calculation time is reduced and, in certain cases, local minima can be avoided as the shape of the objective function depends on the data considered. This technique and its variants are often used to train deep neural networks [e.g., Kingma and Ba, 2017].

In the following, we describe the new aspects introduced to crosshole GPR FWI, which are: (1) The OT distance used to compare GPR traces, (2) the criterion to determine when to switch from the OT to the LS distance and (3) the numerical calculation of the gradient based on master points and subsequent interpolation. Then we provide an overview over the proposed algorithm, followed by a numerical example, discuss various aspects of this new algorithm and finish with conclusions.

2. Method

2.1. Optimal-transport distance

In order to compare an observed GPR trace with a synthetic GPR trace using an OT distance, both traces need to be transformed such that they feature only positive values and contain the same mass (i.e., integrate to the same value). To achieve this, Sambridge et al. [2022] propose to convert each trace (Fig. 1a) to a 2D pseudo probability density function (Fig. 1b), which they call the fingerprint. To obtain the fingerprint, the distance d_{ij} between each element of a grid determined by the subscripts i and j and the closest point on the trace of the waveform is calculated. Then, the 2D pseudo probability density function p_{ij} is obtained as follows:

$$p_{ij} = \exp\left(-\frac{d_{ij}}{s}\right), \quad (1)$$

with s being a user-defined scaling factor. Subsequently, p_{ij} is normalized to represent a proper probability density function that integrates to one. This fingerprint is calculated for both, the observed trace and the synthetic trace.

Calculating any OT distance between a pair of 2D probability density functions is much more demanding than between 1D probability density functions. Therefore, the fingerprint of Fig. 1b is simplified by

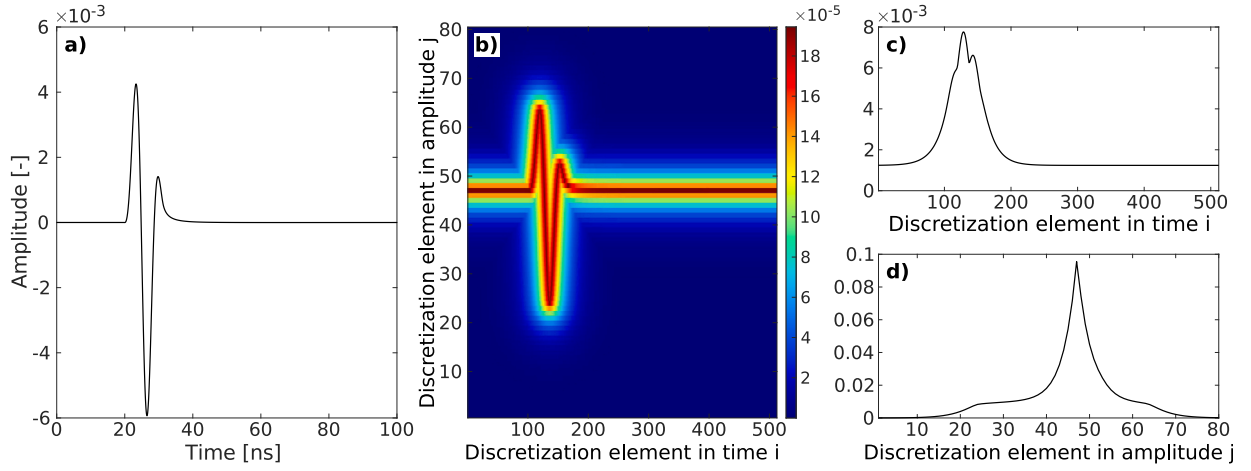


Fig. 1. (a) A GPR waveform, (b) its corresponding normalized fingerprint p_{ij} (2D pseudo probability density function), (c) marginal probability density function p_i^t obtained from integration over amplitude elements and (d) marginal probability density function p_j^u obtained from integration over time elements.

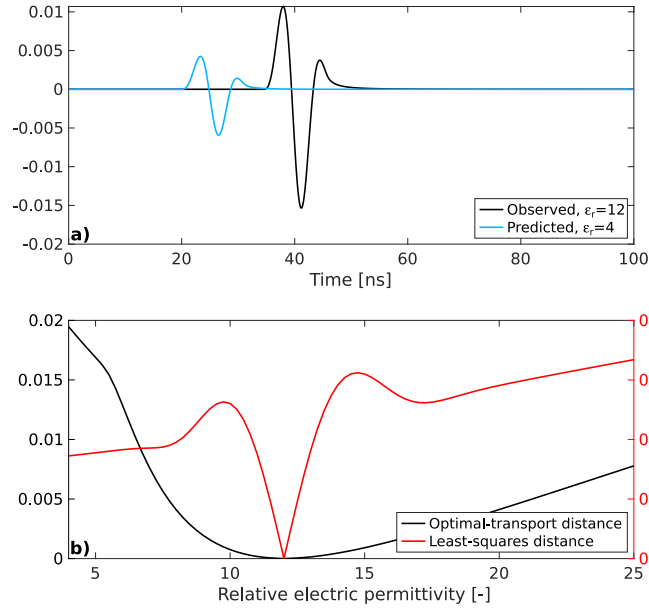


Fig. 2. (a) Observed trace for a relative dielectric permittivity $\epsilon_r = 12$ in black and synthetic trace for $\epsilon_r = 4$ in blue. (b) OT distance W in black and LS distance L in red between the black trace of (a) and a synthetic trace for a range of relative dielectric permittivities. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

converting it to two 1D marginal probability density functions: (1) by integrating over the amplitude elements to obtain a marginal probability density function in time p_i^t (Fig. 1c) and (2) by integrating over the time elements to obtain a marginal probability density function in amplitude p_j^u (Fig. 1d). Thus, the problem of comparing one pair of 2D probability density functions is reduced to comparing two pairs of 1D probability density functions.

Next, the inverse cumulative distribution functions $(P_i^t)^{-1}$ and $(P_j^u)^{-1}$ are calculated from these marginal probability density functions. As we have cumulative distribution functions in time and in amplitude both for the forward-simulated and the observed traces, we can compute

the order-2 Wasserstein distances, W^t and W^u :

$$W^t = \sum_i \left| (P_{i,\text{syn}}^t)^{-1} - (P_{i,\text{obs}}^t)^{-1} \right|^2, \quad (2)$$

$$W^u = \sum_j \left| (P_{j,\text{syn}}^u)^{-1} - (P_{j,\text{obs}}^u)^{-1} \right|^2, \quad (3)$$

where the subscript “syn” and “obs” refer to forward-simulated synthetic data and observed data, respectively. The resulting W^t and W^u are then combined into one distance measure W by a simple addition:

$$W = W^t + W^u. \quad (4)$$

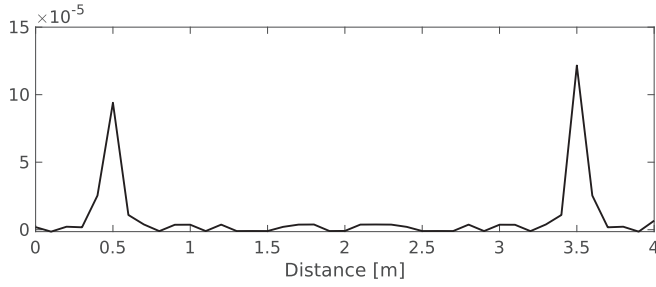


Fig. 3. Gradient of the OT-distance between data based on the homogeneous starting model and the true model shown in Fig. 4a and Fig. 4b, respectively, at a depth of 3 m. The source antenna is located at 0.5 m and the receiver antenna at 3.5 m horizontal distance.

For details of the derivation and implementation of W , we refer to Sambridge et al. [2022].

The LS distance L between a forward-simulated trace $\mathbf{u}_{\text{syn}}$ and an observed trace $\mathbf{u}_{\text{obs}}$ is given by

$$L = \sum_i (u_{i,\text{syn}} - u_{i,\text{obs}})^2. \quad (5)$$

To visually compare the OT distance W to the LS distance L , we present a simple example. It consists of a homogeneous subsurface characterized by a relative dielectric permittivity ϵ_r of 12. Two boreholes at a distance of 3 m from each other penetrate this subsurface. The source antenna of a GPR system is in one borehole at 3 m depth, the receiver antenna in the other borehole is also at 3 m depth. The trace measured in this situation at the receiver is shown as a black curve in Fig. 2a along with a synthetic trace corresponding to a homogeneous subsurface with $\epsilon_r = 4$ (blue curve).

The OT and LS distances between the observed trace for $\epsilon_r = 12$ and a synthetic trace for a homogeneous subsurface with a relative dielectric permittivity between 4 and 25 are shown in Fig. 2b. One can observe that the OT distance (black curve) compared to the LS distance (red curve) has only a global minimum and is much broader. Thus, a gradient-based optimization algorithm would be able to find the global minimum for a starting model at $\epsilon_r = 4$ using the OT distance. This is not the case if using the LS distance, as the algorithm would move in the wrong direction and converge to a local minimum.

2.2. Criterion for switching from the OT to the LS distance

Fig. 2b illustrates that convergence by gradient-based optimization to the global minimum for a starting model far away from the true model is possible if using the OT distance, but not when using the LS distance. However, the global minimum is much better defined for the LS distance, in the sense that the minimum is narrower. Even if the global minimum is mathematically well-defined, the fact that the OT distance provides similar values in a rather wide region around the global minimum makes it more sensitive to simplifications and inaccuracies in implementations of inverse solvers. Also, the error model associated with the LS distance is well defined [e.g., Alumbaugh and Newman, 2000] and the LS distance is much cheaper to compute. Therefore, it is preferable to use the LS distance whenever possible. Consequently, we propose to use the OT distance only to find a model close enough to the global minimum and then switch to the LS distance to find the optimum. This leads to the question of when the algorithm should switch from the OT distance to the LS distance.

For this purpose, we first calculate the time shift t_s between a synthetic and the corresponding observed trace using cross-correlation. We define the relative cross-correlation criterion (RCCC) to be t_s normalized by the automatically-determined dominant period T :

$$RCCC = \frac{t_s}{T}. \quad (6)$$

If the $RCCC$ is smaller than 0.5, the synthetic trace is close enough to the observed trace for a LS inversion to converge to the global minimum. Thus, if enough synthetic traces have $RCCC < 0.5$, the algorithm should switch from the OT distance to the LS distance.

We define N_{RCCC} as the fraction of traces fulfilling this criterion:

$$N_{RCCC} = \frac{\text{traces with } RCCC < 0.5}{\text{all traces}}. \quad (7)$$

The user-defined switching criterion C_s is the critical fraction of traces necessary to switch from the OT distance to the LS distance. Its value lies between 0 and 1. If $N_{RCCC} > C_s$, the switch from the OT to the LS distance is triggered.

2.3. Gradient calculation

To solve the inversion problem with a stochastic gradient descent algorithm, we need the gradient of the OT distance or the LS distance with respect to the model parameters, in our case the relative dielectric permittivity of each model cell. Sambridge et al. [2022] provide analytical expressions for the gradient for their approach of calculating the OT distance. However, in this study, we calculate the gradients explicitly by perturbation for several reasons. First, it allows for an easier implementation and debugging and, second, we seek a flexible and straightforward framework to easily test other distance measures than the two discussed here. Third, the gradients for GPR crosshole tomography problems feature very large values at the antenna locations (Fig. 3). If considered as such in the inversion, the algorithm places by far the largest part of the model update at the antenna locations and does only minimal updates in the rest of the model domain, which leads to convergence problems. We aim to find a better way than just zeroing the gradients at the antenna locations, as is done, for example, by Klotzsche et al. [2013] or Zhou et al. [2020]. Calculating the gradient explicitly allows for a natural way of avoiding this problem as outlined below.

Computing the gradient explicitly by perturbing each model parameter individually requires normally to solve the forward problem as many times as there are parameters in the model. As this is computationally costly, we select a set of randomly chosen master points and perturb the corresponding relative dielectric permittivity by a value of 0.1 to obtain the gradient at these locations. The complete gradient is then approximated by cubic interpolation, which exploits that the gradient for two neighboring cells is correlated. This approach is somewhat related to geostatistical inversion methods based on pilot or master points [e.g., RamaRao et al., 1995; Alcolea et al., 2006]. Bias is avoided as the master points are sampled at random at each iteration. With this approach, we avoid the high values of the gradient close to the antennas by not allowing any master point to be placed very close to the antennas. A minimal distance of 0.2 m, corresponding in our case to two model cells, between the antenna locations and the master point has turned out to be sufficient in our simulations. Calculating the gradient in this way introduces regularization, with a similar effect as traditional model regularization aiming for a smooth model [e.g., Alumbaugh and Newman, 2000]. Therefore, no other regularization is used in our algorithm. To the authors knowledge, calculating the gradient using

master points followed by interpolation has not yet been implemented in geophysics. However, smoothing the gradient as a form of regularization has previously been used [e.g., Yang et al., 2018; Nayak, 2021; Rong et al., 2023].

2.4. Step length of model updates

The gradient provides the direction, in which the model should be updated, but the step length needs to be determined, which is here done by an iterative process. First, the current model $\mathbf{m}_{\text{curr}}$ is updated to obtain various proposed models $\mathbf{m}_{\text{prop}}$ by using a set of pre-determined values for the step length S :

Algorithm 1 Algorithm to determine the optimal step length *bestlen*

```

m, g  $\leftarrow$  the current model and gradient
wobs  $\leftarrow$  observed data
minibatch  $\leftarrow$  Sources and receivers used for mini batch
nrefine  $\leftarrow$  Number of refinement steps
ntry  $\leftarrow$  Number of step-length values tested at each iteration
for it = 1 : nrefine do
  if it = 1 then
    step_len_vec  $\leftarrow$  ntry step-length values to test defined by user
  else
    best_el  $\leftarrow$  Number of element of step_len_vec that corresponds to bestlen.
    step_len_vec  $\leftarrow$  ntry step-length values between step_len_vec(best_el−1) and step_len_vec(best_el+1)
  end if
  for il = 1 : length(step_len_vec) do (can be executed in parallel)
    mprop  $\leftarrow$  m−step_len_vec(il) · g (equation 8)
    wpred  $\leftarrow$  Synthetic data based on proposed model mprop [Irving and Knight, 2006]
    if doOT = 1 then
      w(il)  $\leftarrow$  OT distance between wpred and wobs for minibatch (Section 2.1, Sambridge et al. [2022])
    else
      w(il)  $\leftarrow$  LS distance between wpred and wobs for minibatch (equation 5)
    end if
  end for
  bestlen  $\leftarrow$  step length corresponding to min(w)
end for

```

$$\mathbf{m}_{\text{prop}} = \mathbf{m}_{\text{curr}} - S\mathbf{g}, \quad (8)$$

where $\mathbf{g}$ is the gradient. Second, for each proposed model the OT or the LS distance is calculated, depending on which stage the algorithm is. Third, in a refinement step a new set of values for the step length, around the step length that produced the best model (i.e., the proposed model with the smallest distance), is tested. Again, models are proposed and the corresponding distance measure is computed. The details of this algorithm used to determine the step length are presented in Algorithm

1 (Irving and Knight, 2006; Sambridge et al. [2022]). Calculating the synthetic data and the distance measure for various step-length values within a refinement step can be executed in parallel, making this seemingly expensive procedure rather efficient.

2.5. Overview over the algorithm

All the elements discussed in sections 2.1–2.4 can be combined into a single FWI algorithm, which we refer to as the OT-LS-FWI algorithm. For clarity, it is split here into two parts: (1) the initialization part presented as Algorithm 2 (Irving and Knight, 2006; Sambridge et al. [2022]) and (2) the core part presented as Algorithm 3 (Irving and Knight, 2006;

Sambridge et al. [2022]).

In the initialization part (Algorithm 2), the observed data are loaded (variable **wobs**) and all the parameters are set. Furthermore, the OT or LS distance (variable **w**) for the starting model (variable **m**) and the corresponding gradient (variable **g**) are calculated.

Algorithm 2 The initialization part of the OT-LS-FWI algorithm

wobs, **noisestd** $\leftarrow$ observed data and standard deviation of noise in observed data

m $\leftarrow$ starting model of relative dielectric permittivity

doOT $\leftarrow$ 1 (Flag to determine if currently the OT or the LS distance is used.)

pmax $\leftarrow$ Number of master points for gradient calculation

pgrow, **pmaxlim** $\leftarrow$ growth rate and upper limit for number of master points

maxit $\leftarrow$ Maximum number of iterations

minibatch $\leftarrow$ Sources and receivers used for first mini batch

C_s $\leftarrow$ Switching criterion as defined in section 2.2

wpred $\leftarrow$ Synthetic data based on starting model **m** [Irving and Knight, 2006]

if doOT = 1 then

w $\leftarrow$ OT distance between **wpred** and **wobs** for **minibatch** (section 2.1, Sambridge et al. [2022])

else

w $\leftarrow$ LS distance between **wpred** and **wobs** for **minibatch** (equation 5)

end if

master_points $\leftarrow$ Randomly select **pmax** master points

g $\leftarrow$ Compute gradient based on **master_points** and **minibatch** (section 2.3)

wrmse $\leftarrow \sqrt{\text{mean}\left(\left(\frac{\mathbf{wobs}-\mathbf{wpred}}{\mathbf{noisestd}}\right)^2\right)}$

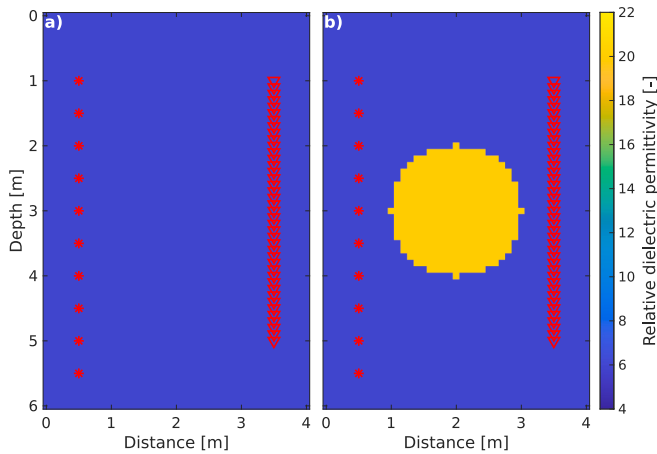


Fig. 4. (a) Homogeneous starting model with a relative dielectric permittivity ϵ_r of 6. (b) True model featuring a large spherical anomaly with $\epsilon_r = 20$. The position of the 10 sources and 41 receivers are indicated by red asterisks and red triangles, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The core part (Algorithm 3) consists of a stochastic gradient descent optimization to iteratively improve the model (variable **m**) by reducing the distance (variable **w**) between the observed and the synthetic data. This is implemented in a while loop that iterates until either the weighted root mean square error (WRMSE) between the observed and the synthetic data is 1.05 or smaller (variable **wrmse**) or the maximum number of iterations (variable **maxit**) is reached. In this while loop, the best step length (variable **bestlen**) is determined according to Algorithm

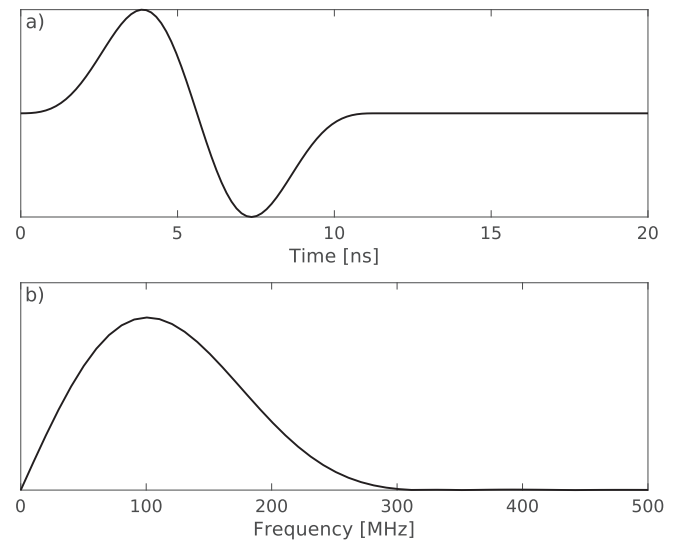


Fig. 5. The source wavelet in (a) the time domain and (b) the frequency domain.

1, the model (variable **m**) is updated and the OT or LS distance (variable **w**) as well as the corresponding gradient (variable **g**) are calculated for the new model. Furthermore, the number of master points for the gradient calculation (variable **pmax**) is increased and the N_{RCC} is calculated at each iteration.

Algorithm 3 The core of the OT-LS-FWI algorithm

```

it ← 1
while it < maxit AND wrmse > 1.05 do
    bestlen ← Optimal step length according to section 2.4
    m ← m − bestlen · g (equation 8)
    wpred ← Synthetic data based on updated model m [Irving and Knight, 2006]
    NRCCC ← fraction of traces with RCCC < 0.5 (section 2.2, eq. 7)
    if NRCCC > Cs then
        doOT ← 0
    end if
    pmax ← pmax · pgrow
    if pmax > pmaxlim then
        pmax ← pmaxlim
    end if
    if doOT = 1 then
        minibatch ← Sources and receivers used for next iteration
        w ← OT distance between wpred and wobs for minibatch (section 2.1, Sambridge et al. [2022])
        master_points ← Randomly select pmax master points
        g ← Compute gradient based on master_points and minibatch (section 2.3)
    else
        w ← LS distance between wpred and wobs for all data
        master_points ← Randomly select pmax master points
        g ← Compute gradient based on master_points and all data (section 2.3)
    end if
    wrmse ←  $\sqrt{\text{mean}\left(\left(\frac{\text{wobs} - \text{wpred}}{\text{noise std}}\right)^2\right)}$ 
    it ← it + 1
end while

```

3. Results

To evaluate our proposed OT-LS-FWI algorithm, we apply it to a synthetic problem that was designed with the goal to be simple yet impossible to solve for the LS inversion with the considered starting model, because this would be the sort of problems for which our proposed algorithm could be useful. The model consists of a large, circular anomaly of a relative dielectric permittivity ϵ_r of 20 embedded in a homogeneous background with $\epsilon_r = 6$ (Fig. 4b). The anomaly could represent, for example, an engineered gallery.

We compare the obtained models with the true subsurface model by calculating the structural similarity index measure (SSIM) introduced by Wang et al. [2003]:

$$\text{SSIM} = \frac{(2\mu_{\text{true}}\mu_{\text{est}} + c_1)(2\sigma_{\text{true,est}} + c_2)}{(\mu_{\text{true}}^2 + \mu_{\text{est}}^2 + c_1)(\sigma_{\text{true}}^2 + \sigma_{\text{est}}^2 + c_2)}, \quad (9)$$

with μ_{true} and μ_{est} being the mean of all pixels of the true and the estimated model, σ_{true} and σ_{est} the standard deviation of all pixels of the true and estimated model and $\sigma_{\text{true,est}}$ the covariance of all pixels between the true and the estimated model. The stabilization constants c_1 and c_2 are set to zero. Values of the SSIM range between -1 and 1 , with a value of 1 indicating that two models are identical.

Two boreholes for the GPR antennas penetrate the model domain at 0.5 and 3.5 m horizontal distance. 10 sources between 1 and 5.5 m depth

are located in the left borehole and 41 receivers between 1 and 5 m depth in the right borehole, resulting in 410 source-receiver pairs. Note that this setup leads to gaps in the ray coverage around the left borehole. This asymmetric experimental design was chosen as it is easy to acquire in the field and it only uses few sources with respect to the number of receivers, thereby, making modeling more efficient. The data are contaminated with Gaussian noise, whose standard deviation is equal to 1 % of the maximum amplitude of the entire dataset. The receivers record traces of a total length of 100 ns with a sample every 0.16 ns. The shape of the source function is shown in Fig. 5.

Meles et al. [2012] showed that a starting model obtained by travel-time inversion does not necessarily lead to a satisfactory convergence of LS-based FWI. To avoid demonstrating this again and for simplicity, we consider in our test case only a homogeneous starting model, which features a relative electric permittivity $\epsilon_r = 6$ (Fig. 4a). The choice of the subsurface model and this starting model ensures, that the corresponding data differ by more than half a period from each other and, thus, being impossible for the LS FWI. The model domain is 6 m deep and 4 m wide. It is discretized in 61 times 41 elements of size 0.1 m times 0.1 m. For the forward modeling, a finer discretization of 0.02 m times 0.02 m is used to ensure numerical stability.

The gradient is calculated at the first iteration by using 64 master points (Variable p_{max} in Algorithms 2 and 3). They consist of the four corners of the domain plus 60 randomly chosen points that are at least 0.2 m away from any GPR antennas. At each further iteration, the

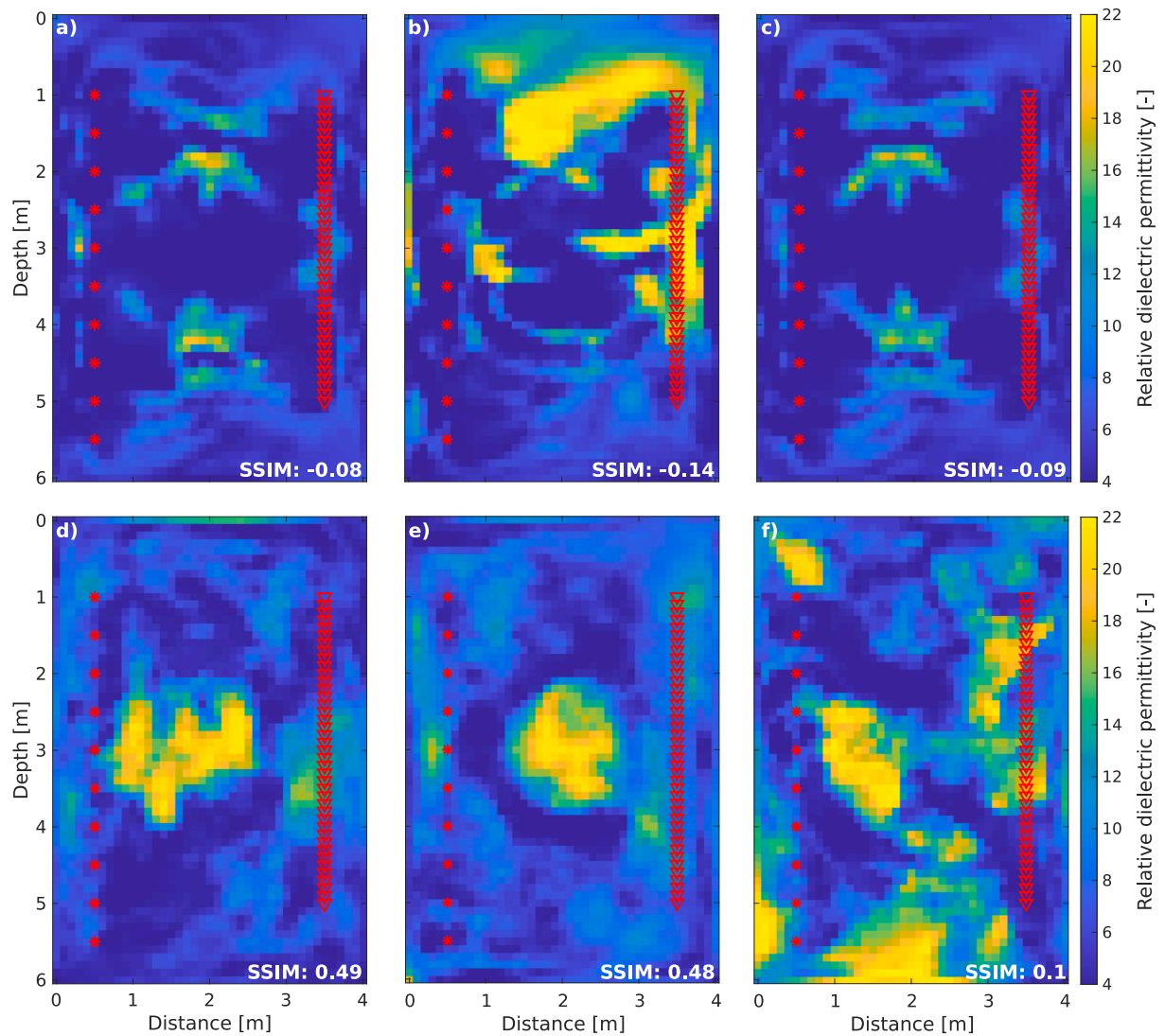


Fig. 6. Inversion results using the LS distance (a-c) and the OT distance (d-f) separately. For each model, the SSIM between the true model (Fig. 4b) and the inversion result is given in the lower right corner.

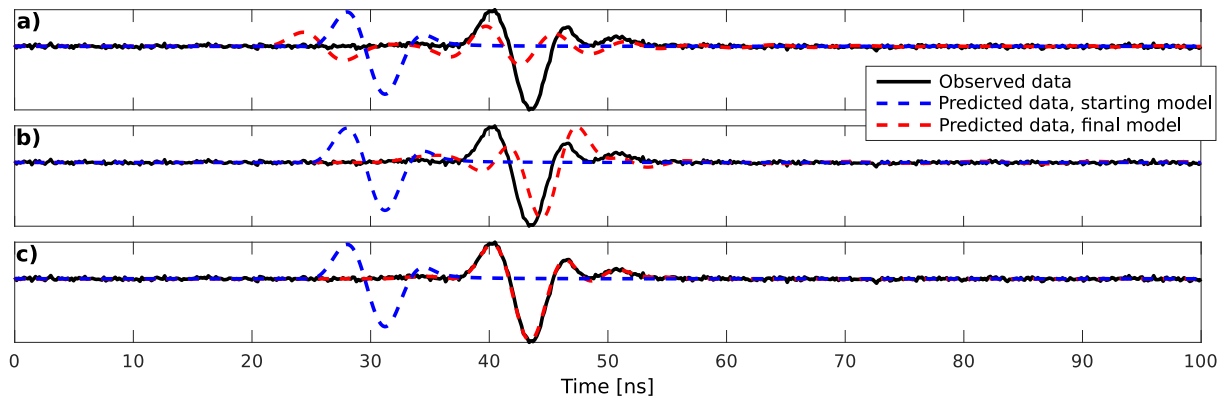


Fig. 7. A trace corresponding to the fourth source and the 21st receiver. The observed data based on the true model (Fig. 4b) is shown in black, the predicted data based on the starting model (Fig. 4a) in blue and the predicted data based on the final model in red for (a) the pure LS inversion (Fig. 6a), (b) the pure OT inversion (Fig. 6d) and (c) the OT-LS-FWI (Fig. 9c).

number of master points is increased by 1.5 % (variable *pgrow*) up to a maximum of 25 % of all model parameters. In our case, 25 % of 2501 model parameters results in a maximum of 625 master points (variable

pmaxlim). The goal behind this choice of parameters is to have a smooth gradient at the start of the inversion that builds the main features of the model. Then, step by step more details are added up to an effective

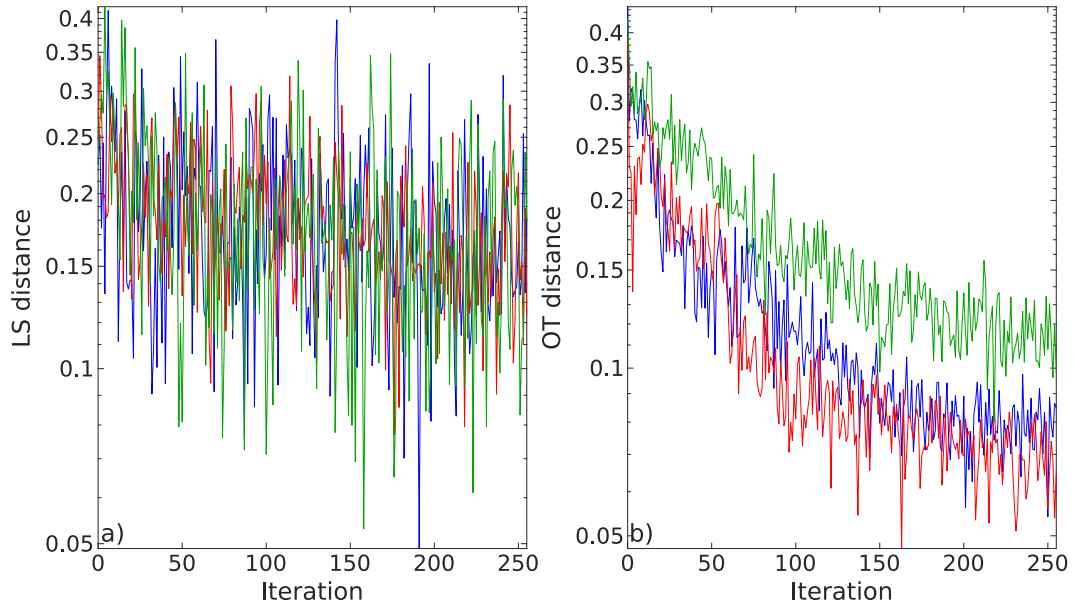


Fig. 8. (a) LS distance and (b) OT distance as a function of iterations for the pure LS or pure OT inversion, respectively. Blue, red and green curves correspond to the inversion runs whose final model is shown in Fig. 6 in the first, second and third column, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

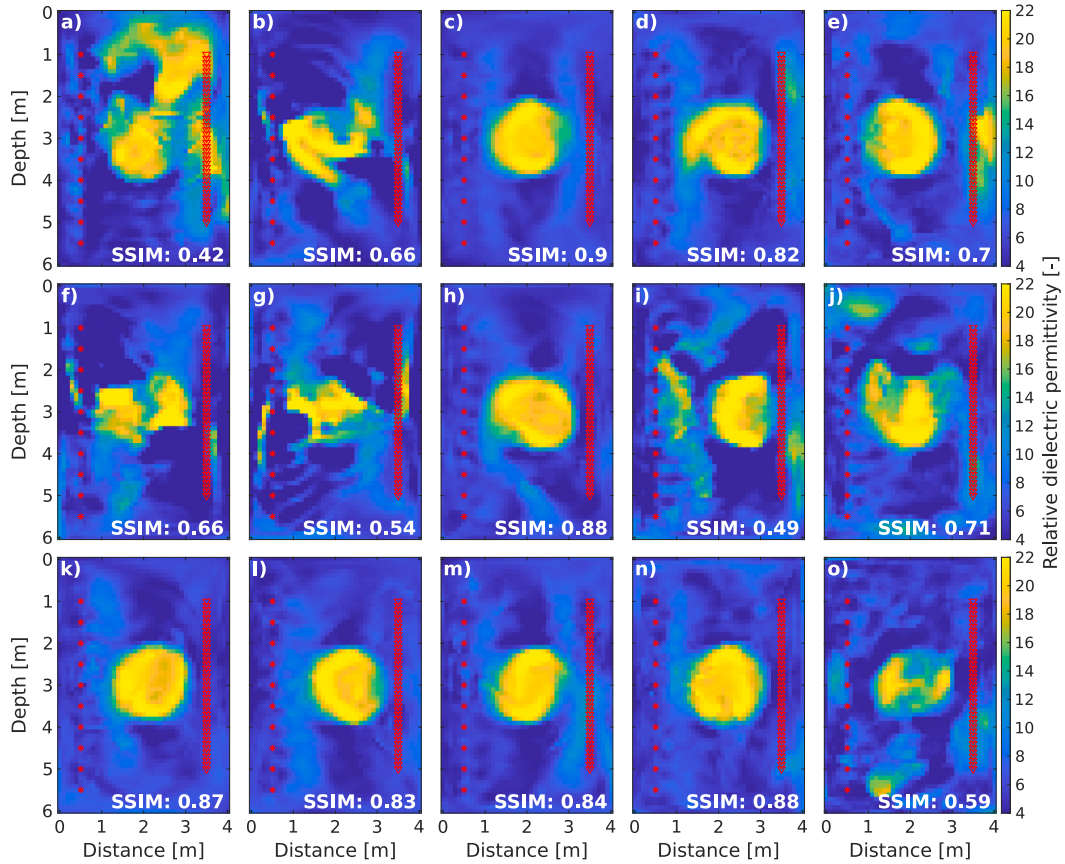


Fig. 9. Subsurface models of relative dielectric permittivity obtained by FWI using first the OT distance followed by the LS distance (OT-LS-FWI). Three runs have been carried out for each value of C_s . First column $C_s = 0.6$, second column $C_s = 0.65$, third column $C_s = 0.7$, fourth column $C_s = 0.75$ and fifth column $C_s = 0.8$. For each model, the SSIM with respect to the true model (Fig. 4b) is given in the lower right corner.

resolution of 0.2 m. As the problem is underdetermined, some form of regularization is necessary, which is achieved here by using a smoothed gradient.

The step length S is determined in three refinement steps (variable $n_{refine} = 3$ in Algorithm 1). In the first refinement step, 16 values (variable $n_{try} = 16$) on a logarithmic scale between 10^{-5} and 10^{10}

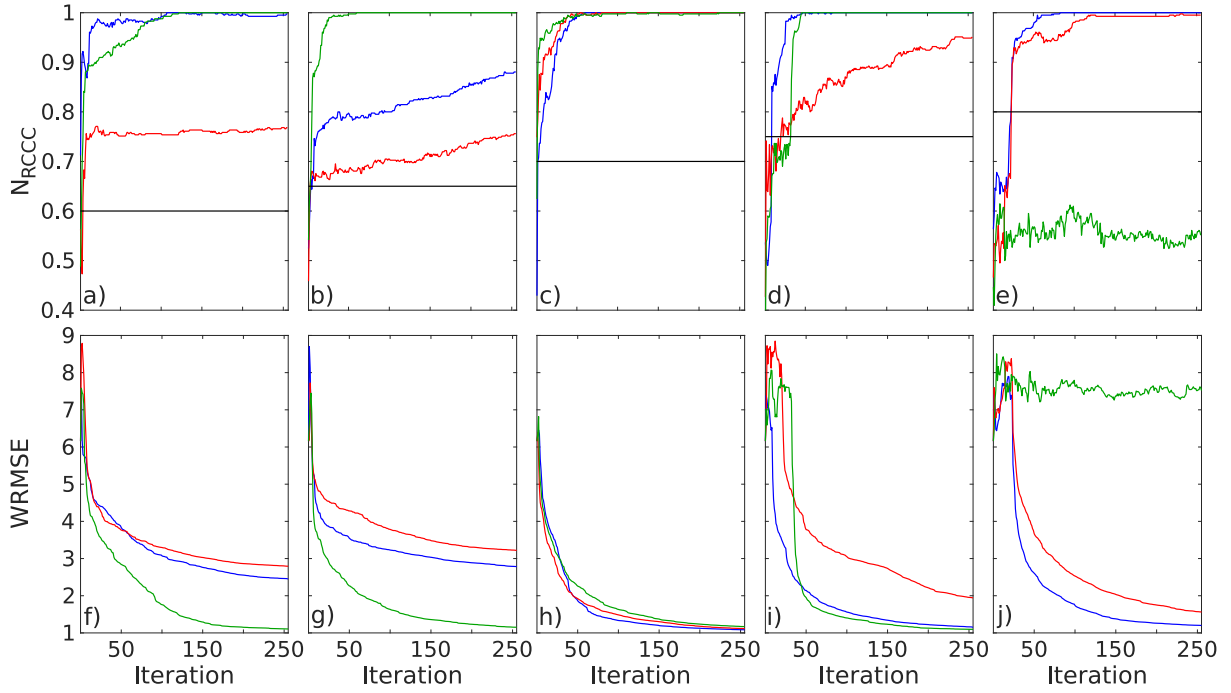


Fig. 10. Top row: N_{RCCC} as a function of iterations for (a) $C_s = 0.6$, (b) $C_s = 0.65$, (c) $C_s = 0.7$, (d) $C_s = 0.75$ and (e) $C_s = 0.8$. The black line indicates the value of C_s in each plot. Bottom row: WRMSE as a function of iterations for (f) $C_s = 0.6$, (g) $C_s = 0.65$, (h) $C_s = 0.7$, (i) $C_s = 0.75$ and (j) $C_s = 0.8$. For all plots, the blue curves correspond to the models in the first row of Fig. 9, the red curves to the second row and the green curves to the third row. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

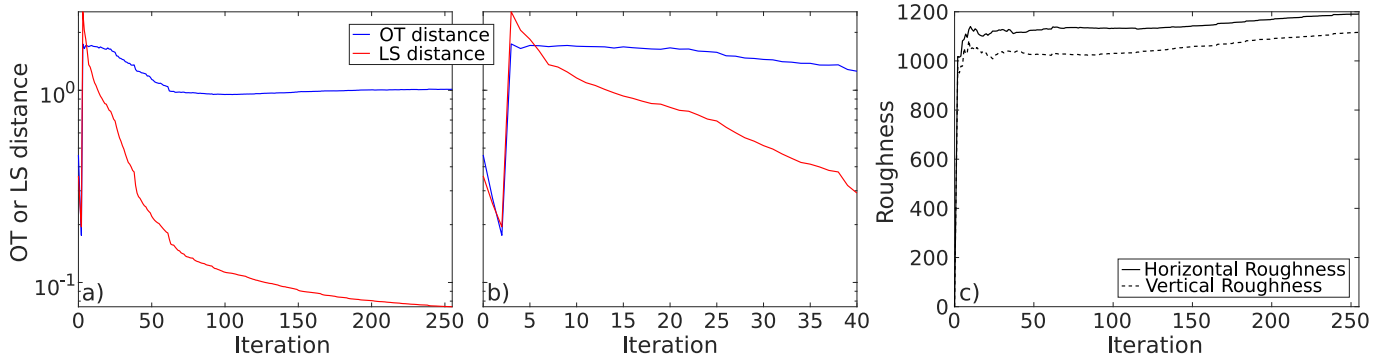


Fig. 11. (a) OT and LS distance as a function of iterations for the OT-LS-FWI algorithm whose final model is shown in Fig. 9c. The switch from OT to LS occurs after iteration 2. (b) Zoom of the first 40 iterations of the plot in (a). (c) Corresponding horizontal and vertical model roughness as a function of iterations.

(variable `step_len_vec`) are tested. In the refinement steps, 16 values around the previously determined best value are tested. As we run the code on 16 cores, all values for the step length are tested simultaneously at each refinement step.

In the stochastic gradient descent algorithm (Algorithm 3), we randomly select at each iteration a subset of the data to calculate the distance between the observed and synthetic data as well as to calculate the gradient. In our case, the subset consists of half of the source positions and a quarter of the receiver positions, resulting in one eighth or 12.5 % of the 410 source-receiver pairs. At every iteration a new random subset is selected.

Running the inversion three times with these settings, with the LS distance only, the algorithm converges each time to a local minimum. Thus, the obtained images of the subsurface (Fig. 6a-c) do not correspond at all to the true subsurface (Fig. 4b), which is also indicated by the negative SSIM values.

This is not a surprise, as the synthetic data corresponding to the starting model are shifted by more than half a period from the observed

data for most source-receiver pairs, resulting in cycle skipping. This can be seen in Fig. 7, which shows the synthetic trace corresponding to the starting model for the fourth source and the 21st receiver with a dashed blue line and the corresponding observed trace with a solid black line. The equivalent synthetic trace corresponding to the final model obtained by the first run of the pure LS inversion (Fig. 6a) is depicted with a dashed red line in Fig. 7a. The algorithm created a model that features peaks in the synthetic trace that align with the observed trace (between 40 and 45 ns) but does not eliminate the peaks present in the synthetic trace corresponding to the starting model (between 25 and 35 ns). Thus, instead of shifting the trace corresponding to the starting model such that the peaks of the synthetic and the observed trace align, the model was adapted such that new peaks were added to the synthetic trace.

The evolution of the LS distance as a function of iterations (Fig. 8a) also shows that the LS inversion is unable to converge to a high-quality model, as the LS distance is on average only slightly reduced during the inversion. Note, that the oscillations in the LS distance are caused by the stochastic gradient descent algorithm. As at each iteration a different

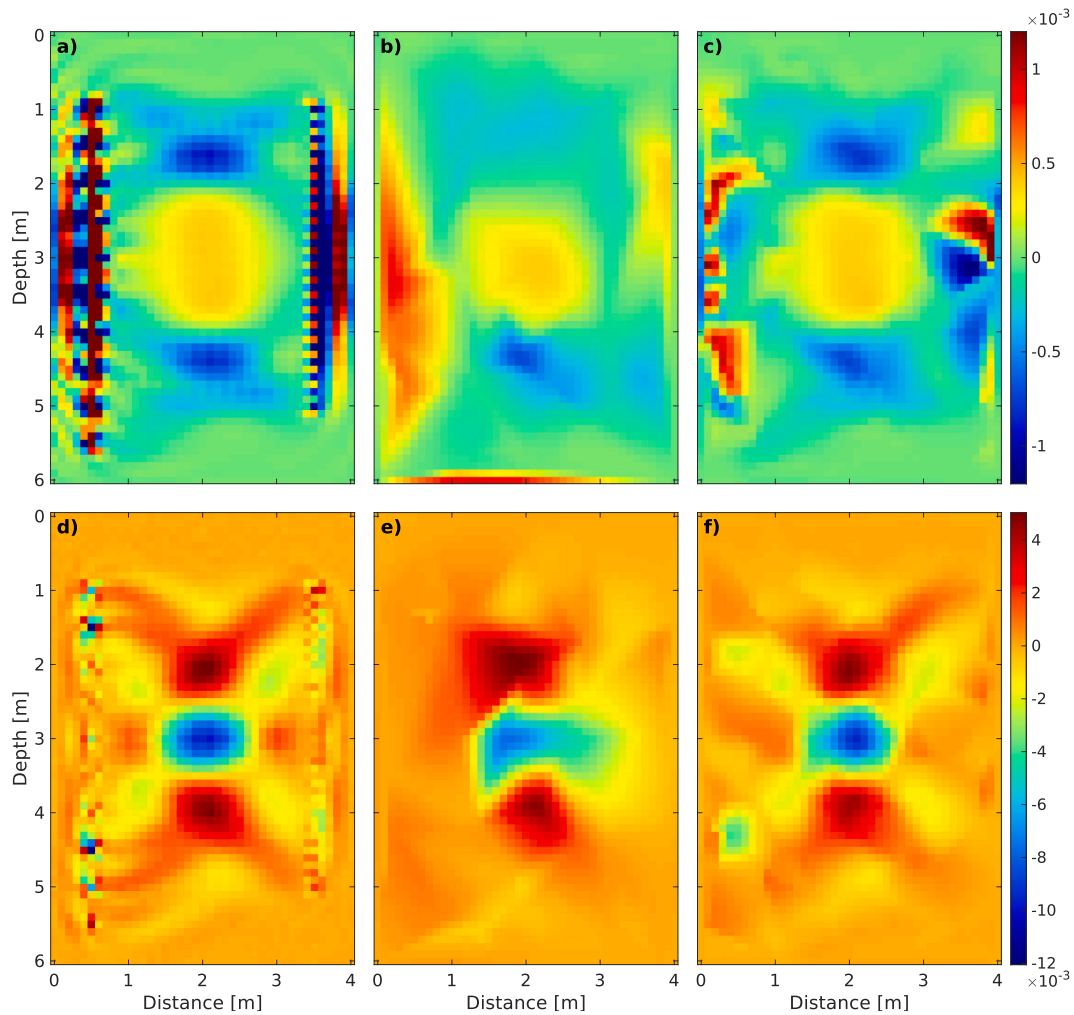


Fig. 12. Gradient of the LS-distance (top row) and the OT-distance (bottom row) with respect to each model parameter (pixel) for the starting model (Fig. 4a) obtained by numerical perturbation of (a, d) each model parameter, (b, e) 64 randomly drawn master points followed by interpolation and (c, f) 256 randomly drawn master points followed by interpolation. The sources are located between 1 and 5.5 m depth at a horizontal distance of 0.5 m. The receivers are located between 1 and 5 m depth at a horizontal distance of 3.5 m.

part of the data are considered, the distance between the observed and the synthetic data might increase from one iteration to the next. The WRMSE for all three runs does not decrease to a value close to one at the end of the inversion, but remains above a value of four further confirming that convergence to a high-quality model did not occur.

The pure OT inversion performs better in the sense that two out of the three example runs obtain a model that bears resemblance with the true model (Fig. 6d-e), which is also indicated by SSIM values of 0.48 and 0.49, respectively, whereas one run does not converge at all (Fig. 6f). Comparing the synthetic trace corresponding to the final model shown in Fig. 6d with the observed trace confirms that the pure OT inversion indeed fared better than the pure LS inversion in the sense that the synthetic trace contains only two major peaks (Fig. 7b) as is also the case for the observed data and not four peaks as was the case for the LS inversion. Thus, the OT inversion was able to adapt the model such that the trace corresponding to the starting model was shifted to fit the observed trace. However, quite a large discrepancy between the synthetic trace and the observed trace remains after the inversion. While the LS inversion was unable to decrease the distance between the observed and the synthetic data significantly, the OT inversion is decreasing the distance by a factor of three or more (Fig. 8b), confirming that the OT inversion is performing better than the LS inversion for this problem. However, the WRMSE, that is inherently a least-squares measure, is with values between 5 and 9 higher than for the pure LS inversion.

The OT inversion is able to move from the starting model towards a reasonable model, but is then unable to improve the details of the model. These results suggest that first using the OT distance to progress towards a model close to the true model and subsequently switching to the LS distance to fill in the details, might provide even better results. Our OT-LS-FWI algorithm is applied next to the same problem. As before, we use half of the source positions and a quarter of the receiver positions to calculate the OT distance and the gradient, but use all the data after the algorithm has switched to the LS distance, because (1) filling in the details of a model works best if all the data are taken into account and (2) jumping out of local minima is no longer required as the inversion is supposed to have moved into the cone of attraction of the global minimum during the first part of the inversion.

Switching from the OT distance to the LS distance occurs if N_{RCC} (Eq. 7) is larger than C_s . Here we test the following values for the switching criterion C_s : 0.6, 0.65, 0.7, 0.75 and 0.8. To avoid faulty conclusions due to the stochastic features of the algorithm, namely (1) a random subset of data is used at each iteration to calculate the OT distance as well as the gradient and (2) random sets of master points are used for the gradient calculation, each case is run three times.

If the switching criterion C_s is smaller than 0.7 (first two columns of Fig. 9), in most cases the algorithm does not converge to a model close to the true subsurface model (Fig. 4b). The SSIM for these models ranges between 0.42 and 0.66, which is comparable to the pure OT inversion

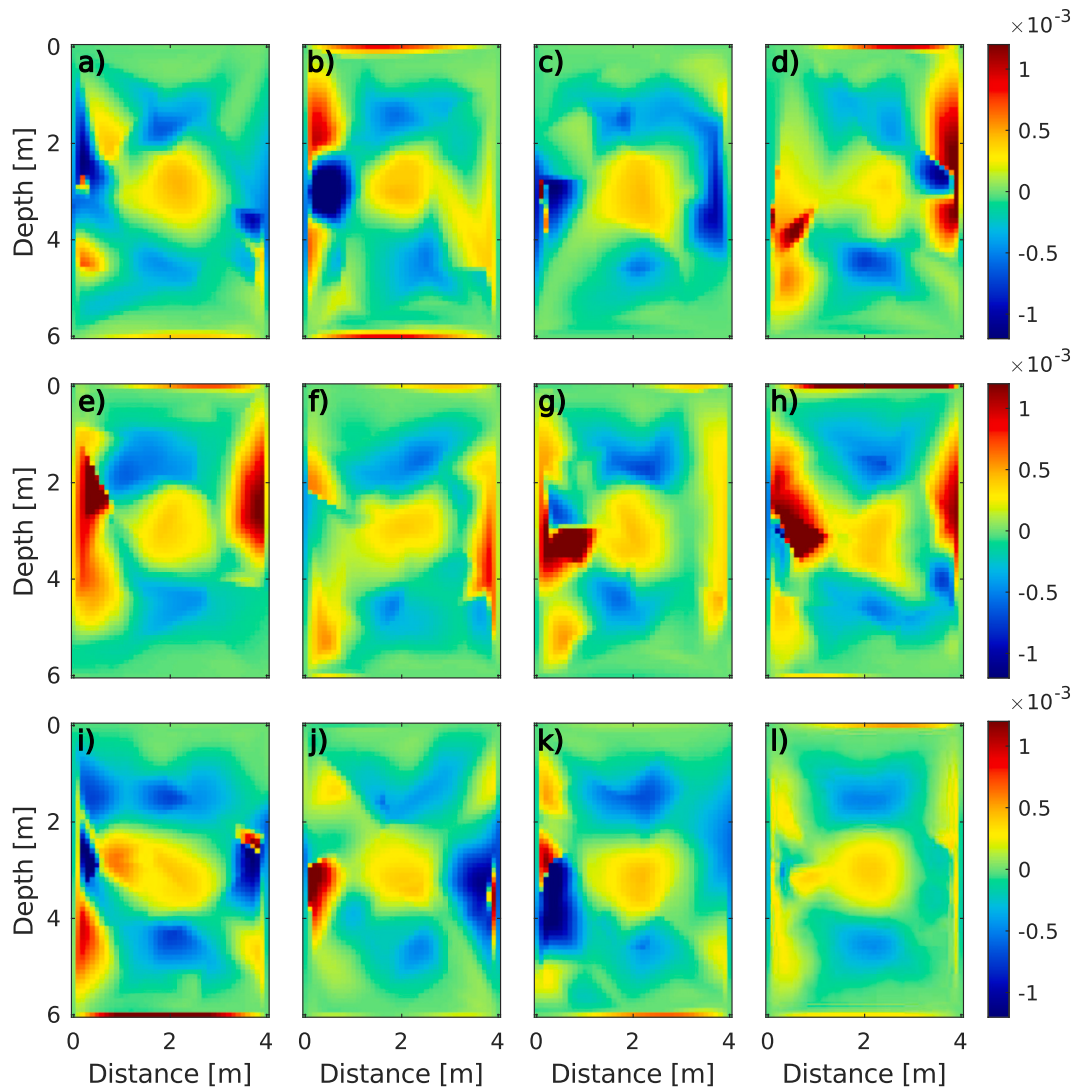


Fig. 13. (a–k) 11 different realizations of the gradient using 64 randomly chosen master points (c.f., Fig. 12b). (l) The mean of these 11 realizations.

results. The exceptions are the third run for $C_s = 0.6$ (Fig. 9k) and the third run for $C_s = 0.65$ (Fig. 9l) with SSIM values of 0.87 and 0.83, respectively. Thus, for $C_s < 0.7$, the fraction of traces that are shifted by less than half a period relative to the corresponding observed traces was for most of the runs too small when the algorithm switched to the LS distance. Therefore, cycle skipping occurred for these runs.

For cases where C_s is equal to or larger than 0.7, a model close to the true subsurface was found in most cases (third, fourth and fifth columns of Fig. 9). The SSIM values for these models generally lie between 0.71 and 0.9. However, there are two exceptions: one model for $C_s = 0.75$ (Fig. 9i, SSIM of 0.49) and one model for $C_s = 0.8$ (Fig. 9o, SSIM of 0.59) are clearly inferior compared to the other models obtained with $C_s \geq 0.7$. However, even for these cases the inversion results are of higher quality compared to the models obtained by the inversion using only the LS distance (Fig. 6a–c).

In general, the models shown in Fig. 9 show no clear imprint of the asymmetric illumination due to the sparse distribution of antennas in the left borehole and the much denser distribution of antennas in the right borehole. For some models (e.g., Figs. 9h or 9i) the anomaly seems to be shifted towards the right, that is the receiver borehole, where the ray coverage is denser. However, other models appear to be symmetric (e.g., Figs. 9c or 9m).

The top row of plots in Fig. 10 shows the N_{RCCC} as a function of iterations for all the runs depicted in Fig. 9. Only two of the runs with $C_s <$

0.7 reached $N_{RCCC} = 1$ (green curves in Fig. 10a and b), implying that for most of these runs some traces showed a shift larger than half a period between the synthetic trace and their observed counterpart even by the end of the inversion. In short, the obtained model is unable to reproduce the observed data adequately.

For $C_s = 0.7$ all runs ended with $N_{RCCC} = 1$ and for $C_s = 0.75$ one run got stuck at $N_{RCCC} = 0.95$ (red curve in Fig. 10d). For $C_s = 0.8$ one run never switched to the LS part of the inversion (green curve in Fig. 10e), because in this case N_{RCCC} was never larger than C_s . Thus, if C_s is chosen too large, the algorithm might never switch to the LS distance. In this case, a region close to the true model has been found, but the optimum has not been located. Thus, the optimum value for C_s lies around 0.7 and 0.75 for the considered test example.

The bottom row of plots in Fig. 10 shows the WRMSE as a function of iterations for the same runs. The WRMSE evolution is roughly a mirror of the N_{RCCC} in the sense that if the N_{RCCC} does not reach a value of 1, the corresponding WRMSE does not decrease to 1 either. On the other hand, if the N_{RCCC} reaches a value of 1, the corresponding WRMSE decrease to almost 1 towards the end of the inversion. Note that the stopping criteria of a WRMSE equal to 1.05 is never reached by any of the example runs shown here, but is likely to be the case if the number of iterations were further increased, because the WRMSE is still gradually improving at the end of the inversion.

Fig. 10 clearly illustrates the value of N_{RCCC} and WRMSE as

diagnostic tools, because all inversions that produced a good final model feature $N_{RCCC} = 1$ and a WRMSSE only slightly above 1 and all inversions that got stuck in a local minimum or did not converge to the bottom of the global minimum feature N_{RCCC} smaller than 1 and a WRMSSE much larger than 1. Thus, in practice only results with $N_{RCCC} = 1$ and WRMSSE close to 1 should be retained.

We can also assess the quality of the OT-LS-FWI algorithm by comparing an observed trace with the corresponding synthetic trace for the starting model and the final model. Fig. 7c shows this comparison for the first run with $C_s = 0.7$ (Fig. 9c) for the fourth source and the 21st receiver. As the observed trace (solid black line) and the synthetic trace based on the final model (dashed red line) overlap perfectly, the figure confirms that the algorithm is able to find a model explaining the data despite the large shift between the observed trace and the synthetic trace based on the starting model (dashed blue line).

For the same inversion run (Fig. 9c), the OT and LS distances are shown as a function of iterations in Fig. 11a and a zoom of the first 40 iterations in Fig. 11b. Here, at iteration zero the distance measures are shown for the starting model. In this example run, the algorithm switched already after two iterations from the OT distance to the LS distance. Thus, for the first two iterations, the inversion is driven by the OT distance (blue curve) and for the remainder by the LS distance (red curve). The switch from the OT to the LS distance is accompanied by an increase of the distance measures, because in the OT phase at each iteration only one eighth of the source-receiver pairs is considered while during the LS phase all source-receiver pairs are taken into account.

Fig. 11c shows the horizontal and vertical roughness of the obtained model as a function of iterations, with roughness defined as the summed absolute difference between neighboring model cells either in the horizontal or in the vertical direction. It can be observed that the roughness increases strongly during the two OT iterations, during which the general structure is obtained and then only increases slightly during the LS part of the inversion, when the details of the model are filled in.

The initial gradients of the LS and the OT distance for this problem are shown in Fig. 12a and d, respectively, if all model parameters (i.e., all pixels of the subsurface image) are perturbed. Strong values of alternating signs are found around the antenna locations at horizontal distances of 0.5 and 3.5 m, which makes convergence of the inversion process very challenging. In order to avoid this issue, we calculate the gradient only at the master points and interpolate to obtain the gradient at the remaining pixels. The corresponding gradients using 64 (Fig. 12b and e) and 256 (Fig. 12c and f) master points show that the strong values and the oscillatory behavior close to the antenna positions largely disappear when using master points to calculate the gradient. Note that the asymmetry of the gradients stems from the random distribution of the master points. It decreases if more points are added.

While the OT gradients shown in Fig. 12e and f are free of significant artifacts, the LS gradients shown in Fig. 12b and c contain artifacts in the neighborhood of the antenna positions. These artifacts, however, are of much smaller amplitude than the antenna imprint seen in Fig. 12a and, therefore, have a smaller impact on the updated model at each iteration. Additionally, as the positions of the master points are changed randomly at each iteration, the artifacts also change at each iteration. Consequently, after multiple iterations the effects of these artifacts on the inverted model average out. This effect is demonstrated in Fig. 13, which shows eleven realizations of the gradient shown in Fig. 12b based on different master points (Figs. 13a–13k). Calculating the average of these eleven realizations reduces these artifacts (Fig. 13l). The same effect takes place by running the inversion for multiple iterations. Therefore, we consider the remaining artifacts in Fig. 12b and c as negligible.

4. Discussion

In this study, we initialize the FWI with a homogeneous model instead of, as is commonly done, with a starting model obtained by travel-time inversion [e.g., Klotzsche et al., 2012]. Our goal is to avoid

this step for the following two reasons: (1) It requires additional user interaction and effort to set the parameters and run the travel-time inversion; (2) a travel-time inversion does not guarantee to find a model sufficiently close to the true model for a successful application of FWI based on the LS distance, for example, in the presence of a strong low-velocity zone. In contrast, the proposed OT-LS-FWI allows to obtain a detailed subsurface model by running one algorithm and without the need to find first a good starting model.

One of the main features of the OT-LS-FWI algorithm (Algorithms 2 and 3) is the usage of the OT distance during the first few iterations of the inversion before switching to the LS distance. We have shown that the cone of attraction of the global minimum is much broader for the OT distance than for the LS distance (Fig. 2b). This has the advantage that the dependence of the starting model can be reduced as the global minimum can be found with almost any starting model, in our case a homogeneous subsurface model. The disadvantage of the OT distance, however, is that the valley surrounding the global minimum does not feature a large gradient. In a perfect noiseless world, this would not hinder convergence to the optimum. However, in the presence of noise in the data and in connection with our usage of an imperfect gradient, which is a numerical approximation of the true gradient based only on a fraction of the data, the global minimum can become essentially flat, making convergence to the optimum difficult. Additionally, by considering a different random subset of the data at each iteration, the position of the optimum is altered, which is not a problem at the beginning of the inversion, because the rough direction stays the same, but hinders convergence close to the global minimum at the end of the iterative process. Using an exact gradient [as provided by Sambridge et al., 2022] instead of the explicit calculation used here, will probably reduce this convergence issue somewhat, but it is not expected to disappear if a stochastic gradient descent method is used and data noise is present. It is, thus, difficult to find the optimal combination of model parameters using only the OT distance.

Deng et al. [2022] observed the same problem in their electromagnetic OT FWI. They proposed a hierarchical approach when converting the recordings of the electromagnetic waves into probability density functions. In the early stage, the conversion was carried out such that no local minima appeared and the global minimum was broad. In the later stage, a different value for a conversion parameter produced probability density functions that lead to a more narrow global minimum with local minima present as well. Here, we propose a comparable two-stage approach, which uses first the OT distance to obtain an objective function featuring a single broad minimum and no local minima and then to switch to the LS distance once the current model is within the cone of attraction of the much narrower global minimum of the LS distance. While Deng et al. [2022] exploited the conversion from the waveform to the probability density function to obtain favorable convergence properties, we use two different distance measures to reach the same goal. Our approach has the advantage that the expensive-to-calculate OT distance needs to be calculated only in the first few iterations of the algorithm, while the algorithm of Deng et al. [2022] requires it to be calculated at all iterations.

Note that the narrow global minimum of the LS distance carries both, advantages and disadvantages. At the beginning of the inversion, the LS distance is a bane and needs to be avoided as it is difficult to reach the global minimum. However, once a model within the cone of attraction of the LS distance has been found, the LS distance becomes a boon, because its narrow global minimum prevents the algorithm from diverging even when the problem features many unknowns and only limited model regularization. The fact that the steep gradient around the global minimum in the LS distance stabilizes the algorithm, offers an additional form of model regularization.

A crucial choice is when to switch from the OT to the LS distance. For this reason, we defined the switching criterion $N_{RCCC} > C_s$, with N_{RCCC} defined in Eq. 7 and C_s a user-defined parameter. If the user chooses a too small value for C_s , the algorithm will switch too early to the LS

distance and will get stuck in a local minimum because of cycle skipping. If the value chosen for C_s is too large, the algorithm might never switch, because the switching criterion is never reached. The range of values for C_s , for which our example could be solved in a majority of cases, is found to lie between 0.7 and 0.8 (Fig. 9). It is expected that it will not change strongly for other problems, because the normalization with the dominant period T (Eq. 6) and the number of traces (Eq. 7) makes the measure independent of the frequency and the size of the data set. We recommend a value of 0.7, as it empirically leads to the best results. This value should be decreased if switching does not occur or increased if the N_{RCC} and the WRMSE do not approach a value of 1 (Fig. 10). We only recommend to use inversion results for which the final N_{RCC} is 1 and the WRMSE is close to the pre-defined stopping criterion. If one run does not converge it is possible to rerun the algorithm as its stochastic features lead to slightly different results and convergence characteristics each time.

Another defining feature of the proposed algorithm is the way the gradient is obtained. Calculating the gradient explicitly is much more computationally expensive than, for example, using the adjoint-state method [e.g., Plessix, 2006]. However, as the calculation of the gradient of the objective function with respect to one model parameter is independent of the calculation of the gradient with respect to another model parameter, the gradient can be calculated in parallel. As our approach based on gradient interpolation does not require to calculate the gradient for each model parameter, and as nowadays massively parallel computational hardware is available, the computational disadvantage of our gradient calculation becomes a minor issue. Our gradient calculation leads to three advantages. First, by avoiding to place master points close to the antenna locations, we do not suffer from extreme gradient values at these positions, which required previous investigators to avoid updating the model close to the boreholes. Second, we intrinsically smooth the model update and with that the model itself, without needing an explicit model-regularization term in the objective function. The number of master points can be seen as the regularization parameter, which is directly linked to an effective resolution of the obtained image. Here, we have chosen to have a maximum of a quarter of the pixels to be master points, which resulted in an effective resolution of 0.2 m. Third, this framework makes it easy to implement and test new distance measures, as only the function calculating the distance between two waveforms needs to be replaced, but not the methodology behind the gradient calculation.

To avoid the strong values of the gradient close to the antenna locations, a user-defined parameter is introduced. This is the value of the critical distance between the antenna locations and possible master-point locations. Its optimal choice depends on the wavelength of the GPR signal and, thus, on the frequency of the signal and the material of the subsurface. The size of this critical distance might also depend on the distance measure, as indicated by Fig. 12. While the gradient appears to be free of strong values close to the antenna locations for the OT distance (Fig. 12f), some imprint remains in the gradient of the LS distance (Fig. 12c). However, by applying at each iteration a gradient that is based on a new set of randomly chosen master points, these artifacts are different at each iteration and interfere destructively when running the inversion over multiple iterations (Fig. 13 l). In this study, we have chosen a value of 0.2 m (two inversion grid cells) regardless of the distance measure.

Note that circumventing the strong values of the gradient using a preconditioner [Kurzmann et al., 2013] also requires a user-defined parameter. In that case, it is a stabilization parameter, which is less intuitive to choose than the critical distance that needs to be defined in our approach. On the other hand, the choice of the stabilization parameter of the preconditioner has only a minor impact on the final result [Klotzsche et al., 2019]. Computationally, the preconditioner is much more efficient to calculate than the master-point approach, but that is an advantage that disappears when massively parallel hardware is available. The large advantages of the master-point approach are, as

mentioned above, that it provides also an intrinsic model regularization without the need of defining a model-regularization term in the objective function and that it allows to incorporate other distance measures in a straightforward way without the need of adapting the gradient calculation.

The third new feature in the context of crosshole GPR FWI is our usage of a stochastic gradient descent algorithm, which only considers a fraction of the data at each iteration. The main benefit is a reduction of computational time. As calculating the OT distance is more expensive than calculating the LS distance, we use the stochastic gradient descent algorithm only during the OT phase of the OT-LS-FWI algorithm and switch to the standard form of gradient descent in the LS phase. Thus, during the OT phase we used one eighth of the available data at each iteration. Adequate results can probably be obtained by a further reduction of the amount of data considered at each iteration, which would lead to a further speed-up. However, it might also lead to more iterations, which would counter-balance the speed-up gained by the data reduction.

The runtime of an OT inversion step using one eighth of the data takes 50 % more time than an LS inversion step using all the data for the same amount of master points for the gradient. Indeed, the additionally required computational time for an OT inversion step is to a large part compensated by only considering 12.5 % of the data. Considering the runtime of the whole inversion, the additional runtime for the individual OT inversion steps does not contribute significantly to the overall runtime of our inversion, because the OT inversion steps are done for a few iterations at the beginning of the inversion process, when only a small number of master points are used for the calculation of the gradient and, thus, cost less computational time. For example, for $C_s = 0.7$ a maximum of three OT iterations were necessary in the worst case to fulfill the switching criterion. These three iterations took 18 min (i.e., 6 min per iteration) on our dated computer cluster using 16 “AMD Opteron 6136” processors at 0.8 GHz. As only a few OT iterations are necessary, there is no significant computational gain from performing a travel-time inversion prior to our OT-LS-FWI. However, if the switching criterion C_s is set higher, more OT iterations are performed, which can seriously increase the computational cost of the proposed OT-LS-FWI. In the worst case, the algorithm never switches to the more efficient LS inversion steps. This illustrates the importance of C_s with respect to the computational costs of the algorithm.

The runtime of the explicit gradient-calculation is the runtime of one distance-measure evaluation (including solving the forward problem) times the number of master points, because for each master point one distance-measure evaluation is required. If run in parallel, this runtime can be divided by the number of CPUs used, assuming that the number of master points is an integer multiple of the number of CPUs used.

In this study, we assumed the source wavelet to be known. This is not the case if field data are considered and, therefore, the source wavelet needs to be estimated along with the subsurface model. This could be achieved using the approach by Hunziker et al. [2019] using six master points to define the shape of the source wavelet and one scaling parameter to define its amplitude. This allows to characterize a rather complicated source wavelet with only 13 unknowns (two for each master point plus one scaling factor). The gradient of these 13 unknowns can easily be calculated within the framework of explicit gradient calculation proposed here.

5. Conclusions

We propose a new gradient-based crosshole GPR FWI algorithm using the OT distance for the first few iterations until a model is found that is close enough to the true subsurface, such that convergence to the global minimum can be ensured using the conventional LS distance. At that point, the algorithm switches to the LS distance, because it is faster to compute and the global minimum is better defined for our implementation. The criterion proposed for switching from one distance

measure to the other is based on a comparison of the time shift of each synthetic trace with its corresponding observed trace. We found that if 70 to 80 % of the synthetic traces are shifted by less than half a period with respect to the corresponding observed traces, FWI using the LS distance converges typically to the global minimum.

The gradient of the objective function with respect to the model parameters is computed explicitly by perturbing a randomly chosen subset of model parameters, so-called master points. The gradient with respect to the remaining model parameters is obtained by interpolation. As we do not allow master points to be placed close to the antenna positions, our gradient does not suffer from extreme values at these locations that often hinder a proper convergence in conventional FWI. Furthermore, depending on the number of master points used, the gradient is smoothed to different degrees and, thus, leads to smooth models without including an explicit model-regularization term in the objective function. It also provides a flexible framework to test various distance measures. The additional computation time is limited if the code is run on massively parallel hardware.

To further decrease the computational time when the algorithm uses the OT distance, we rely on stochastic gradient descent that uses a random subset of the available data, which is changed at each iteration. In our example, using one eighth of the data worked well. However, this reduction could potentially be even larger to further reduce computational time.

As next steps, the estimation of the source wavelet and a conductivity model should be included before applying the algorithm to a more realistic model and eventually to field data. To further improve the efficiency of the algorithm, a Gauss-Newton optimization should be considered instead of the currently implemented gradient-descent approach. Eventually, a resolution analysis and an uncertainty quantification should be developed. It could also be valuable to try our OT-LS-FWI approach on seismic data.

CRedit authorship contribution statement

Jürg Hunziker: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Giovanni Meles:** Methodology, Investigation, Conceptualization. **Niklas Linde:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

We invite interested researchers to further develop the presented algorithm and the associated code: https://github.com/jhunziker/OT_LS_FWI

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¹ <https://github.com/msambridge/waveform-ot>

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